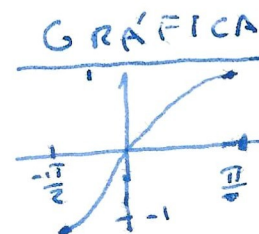


1) FUNCIONES TRIGONOMÉTRICAS

Función
i) $y = \sin(x)$

DOMINIO
 $[-\frac{\pi}{2}, \frac{\pi}{2}]$

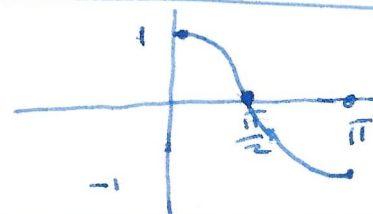
RANGO
 $[-1, 1]$



ii) $y = \cos(x)$

$[0, \pi]$

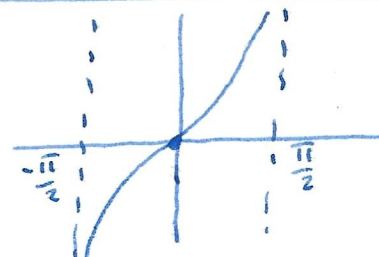
$[-1, 1]$



iii) $y = \tan(x)$

$(-\frac{\pi}{2}, \frac{\pi}{2})$

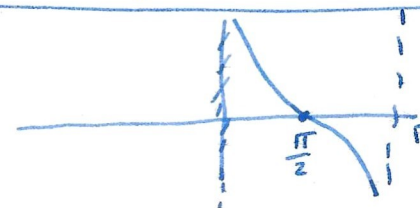
$\mathbb{R}$



iv) $y = \cot(x)$

$(0, \pi)$

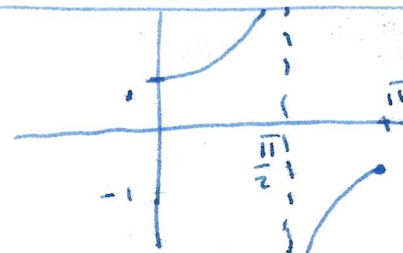
$\mathbb{R}$



v) $y = \sec(x)$

$[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$

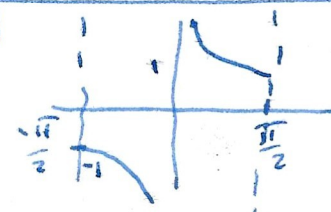
$(-\infty, -1] \cup [1, \infty)$



vi) $y = \csc(x)$

$[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$

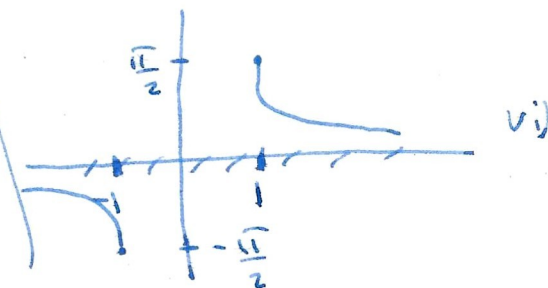
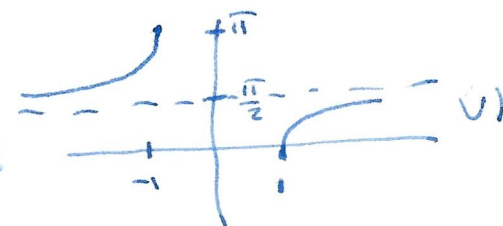
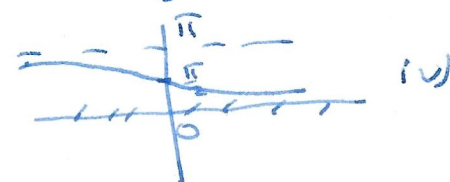
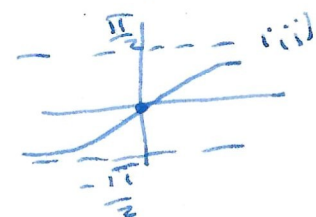
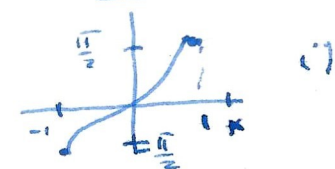
$(-\infty, -1] \cup [1, \infty)$



2) FUNCIONES TRIGONOMÉRICAS INVERSA

<u>Función</u>	<u>Domínio</u>	<u>Rango</u>
i) $y = \sin^{-1}(x)$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
ii) $y = \cos^{-1}(x)$	$[-1, 1]$	$[0, \pi]$
iii) $y = \tan^{-1}(x)$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
iv) $y = \cot^{-1}(x)$	$\mathbb{R}$	$(0, \pi)$
v) $y = \sec^{-1}(x)$	$(-\infty, -1] \cup [1, \infty)$	$[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$
vi) $y = \csc^{-1}(x)$	$(-\infty, -1] \cup [1, \infty)$	$[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$

Grafica



- Función
- i) $y = \sin^{-1}(x)$
- ii) $y = \cos^{-1}(x)$
- iii) $y = \tan^{-1}(x)$

DERIVADA

$\frac{1}{\sqrt{1-x^2}}, |x| < 1$

$-\frac{1}{\sqrt{1-x^2}}, |x| < 1$

$\frac{1}{x^2+1}$

INTEGRAL

$\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C$

$\int \frac{du}{a^2+u^2} = \frac{1}{a} \tan^{-1}\left(\frac{u}{a}\right) + C$

$\int \frac{du}{u \sqrt{u^2-a^2}} = \frac{1}{a} \sec^{-1}\left|\frac{u}{a}\right| + C$

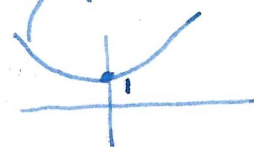
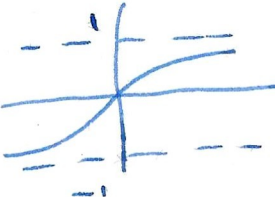
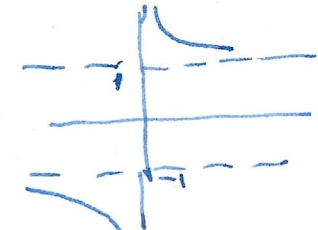
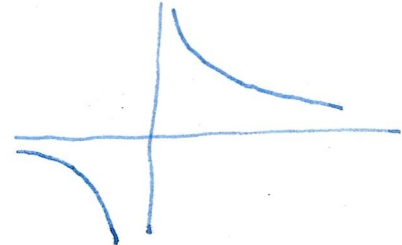
- iv) $y = \cot^{-1}(x)$
- v) $y = \sec^{-1}(x)$
- vi) $y = \csc^{-1}(x)$

$-\frac{1}{x^2+1}$

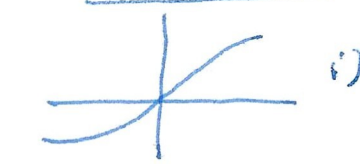
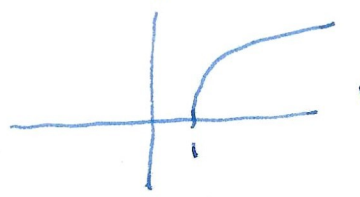
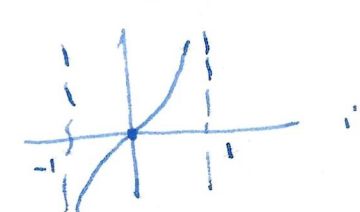
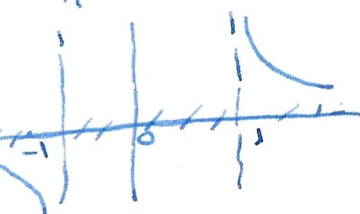
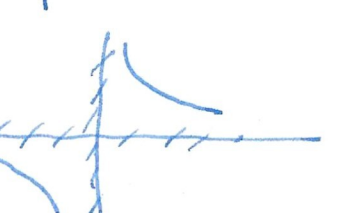
$\frac{1}{|x| \sqrt{x^2-1}}, |x| > 1$

$-\frac{1}{|x| \sqrt{x^2-1}}, |x| > 1$

3) FUNCIONES HIPERBOLICAS

<u>Función</u>	<u>Domínio</u>	<u>Rango</u>	<u>Gráfica</u>
i) $\sinh(x) = \frac{e^x - e^{-x}}{2}$	$\mathbb{R}$	$\mathbb{R}$	
ii) $\cosh(x) = \frac{e^x + e^{-x}}{2}$	$\mathbb{R}$	$[1, \infty)$	
iii) $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	$\mathbb{R}$	$(-1, 1)$	
iv) $\coth(x) = \frac{e^x + e^{-x}}{e^x - e^{-x}}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, -1) \cup (1, \infty)$	
v) $\operatorname{sech}(x) = \frac{1}{\cosh(x)} = \frac{2}{e^x + e^{-x}}$	$\mathbb{R}$	$(0, 1]$	
vi) $\operatorname{csch}(x) = \frac{1}{\sinh(x)} = \frac{2}{e^x - e^{-x}}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$	

4) FUNCIONES HIPERBOLICAS INVERSAS

<u>Función</u>	<u>Domínio</u>	<u>Rango</u>	<u>Gráfica</u>
i) $y = \operatorname{sen}^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	
ii) $y = \operatorname{cos}^{-1} x$	$[-1, 1]$	$[0, \pi]$	
iii) $y = \operatorname{tan}^{-1} x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$	
iv) $y = \operatorname{cot}^{-1} x$	$\mathbb{R}$	$(0, \pi)$	
v) $y = \operatorname{csc}^{-1} x$	$[-\infty, -1] \cup [1, \infty)$	$(-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2})$	
vi) $y = \operatorname{sec}^{-1} x$	$[-\infty, -1] \cup [1, \infty)$	$(-\pi, 0) \cup (0, \pi)$	

(5)

TABLA DE DERIVADAS E INTEGRALES PARA FUNCIONES HIPERBOLICAS

<u>Función</u>	<u>DERIVADA</u>	<u>INTEGRAL</u>
i) $y = \sinh(x)$	$\cosh(x)$	$\int \sinh(x) dx = \cosh(x) + C$
ii) $y = \cosh(x)$	$\sinh(x)$	$\int \cosh(x) dx = \sinh(x) + C$
iii) $y = \tanh(x)$	$\operatorname{sech}^2(x)$	$\int \operatorname{sech}^2(x) dx = \tanh(x) + C$
iv) $y = \coth(x)$	$-\operatorname{csch}^2(x)$	$\int \operatorname{csch}^2(x) dx = -\coth(x) + C$
v) $y = \operatorname{sech}(x)$	$-\operatorname{sech}(x) \tanh(x)$	$\int \operatorname{sech}(x) \tanh(x) dx = -\operatorname{sech}(x) + C$
vi) $y = \operatorname{csch}(x)$	$-\operatorname{csch}(x) \coth(x)$	$\int \operatorname{csch}(x) \coth(x) dx = -\operatorname{csch}(x) + C$

TABLA DE DERIVADAS E INTEGRALES PARA FUNCIONES HIPERBOLICAS INVERSA

<u>Función</u>	<u>DERIVADA</u>	<u>INTEGRAL</u>
i) $y = \sinh^{-1}(x)$	$\frac{1}{\sqrt{1+x^2}}$	$\int \frac{du}{\sqrt{a^2+u^2}} = \sinh^{-1}\left(\frac{u}{a}\right) + C \quad a > 0$
ii) $y = \cosh^{-1}(x)$	$\frac{1}{\sqrt{x^2-1}} \quad x > 1$	$\int \frac{du}{\sqrt{u^2-a^2}} = \cosh^{-1}\left(\frac{u}{a}\right) + C \quad u > a > 0$
iii) $y = \tanh^{-1}(x)$	$\frac{1}{1-x^2} \quad x < 1$	$\int \frac{du}{a^2-u^2} = \begin{cases} \frac{1}{a} \tanh^{-1}\left(\frac{u}{a}\right) + C & u^2 < a^2 \\ \frac{1}{a} \coth^{-1}\left(\frac{u}{a}\right) + C & u^2 > a^2 \end{cases}$
iv) $y = \coth^{-1}(x)$	$\frac{1}{1-x^2} \quad x > 1$	
v) $y = \operatorname{sech}^{-1}(x)$	$-\frac{1}{x\sqrt{1-x^2}} \quad 0 < x < 1$	$\int \frac{du}{u\sqrt{a^2-u^2}} = -\frac{1}{a} \operatorname{sech}^{-1}\left(\frac{u}{a}\right) + C \quad 0 < u < a$
vi) $y = \operatorname{csch}^{-1}(x)$	$-\frac{1}{ x \sqrt{1+x^2}} \quad x \neq 0$	$\int \frac{du}{u\sqrt{a^2+u^2}} = -\frac{1}{a} \operatorname{csch}^{-1}\left(\frac{u}{a}\right) + C \quad u \neq 0 \text{ and } u > 0$

6) ALGUNAS IDENTIDADES IMPORTANTES

- 1) - PARA FUNCIONES TRIGONOMÉTRICAS, VER LA PAGINA DEL CURSO EN CANVAS.
- 2) - PARA FUNCIONES TRIGONOMÉTRICAS INVERSAS:

$$i) \cos^{-1}(x) + \cos^{-1}(-x) = \pi$$

$$ii) \sec^{-1}(x) = \cos^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2} - \sin^{-1}\left(\frac{1}{x}\right)$$

$$iii) \cos^{-1}(x) = \frac{\pi}{2} - \sin^{-1}\left(\frac{1}{x}\right)$$

$$iv) \cot^{-1}(x) = \frac{\pi}{2} - \tan^{-1}(x)$$

$$v) \csc^{-1}(x) = \frac{\pi}{2} - \sec^{-1}(x)$$

- 3) - PARA FUNCIONES HIPERBÓLICAS:

$$i) \cosh^2(x) - \sinh^2(x) = 1$$

$$ii) \sinh(2x) = 2\sinh(x)\cosh(x)$$

$$iii) \cosh(2x) = \cosh^2(x) + \sinh^2(x)$$

$$iv) \cosh^2(x) = \frac{\cosh(2x) + 1}{2}$$

$$v) \sinh^2(x) = \frac{\cosh(2x) - 1}{2}$$

$$vi) \tanh^2(x) = 1 - \operatorname{sech}^2(x)$$

$$vii) \coth^2(x) = 1 + \operatorname{csch}^2(x)$$

- 4) - PARA FUNCIONES HIPERBÓLICAS INVERSAS:

$$i) \operatorname{sech}^{-1}(x) = \cosh^{-1}\left(\frac{1}{x}\right)$$

$$ii) \operatorname{csch}^{-1}(x) = \sinh^{-1}\left(\frac{1}{x}\right)$$

$$iii) \cot^{-1}h(x) = \tanh^{-1}\left(\frac{1}{x}\right)$$

7) PARA EVALUAR FUNCIONES HIPERBÓLICAS INVERSA

$$i) \sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1}) \quad x \in \mathbb{R}$$

$$ii) \cosh^{-1}(x) = \ln(x + \sqrt{x^2 - 1}) \quad x \geq 1$$

$$iii) \tanh^{-1}(x) = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1$$

$$iv) \operatorname{sech}^{-1}(x) = \ln\left(\frac{1 + \sqrt{1-x^2}}{x}\right), \quad 0 < x \leq 1$$

$$v) \operatorname{csch}^{-1}(x) = \ln\left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|}\right), \quad x \neq 0$$

$$vi) \operatorname{coth}^{-1}(x) = \frac{1}{2} \ln\left(\frac{x+1}{x-1}\right), \quad |x| > 1$$